9-W1　直线段的端点坐标

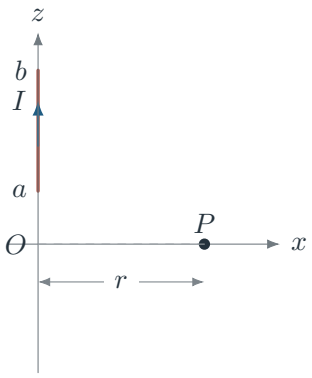


图 9-9　题图　垂足在直线段外，端点坐标同号

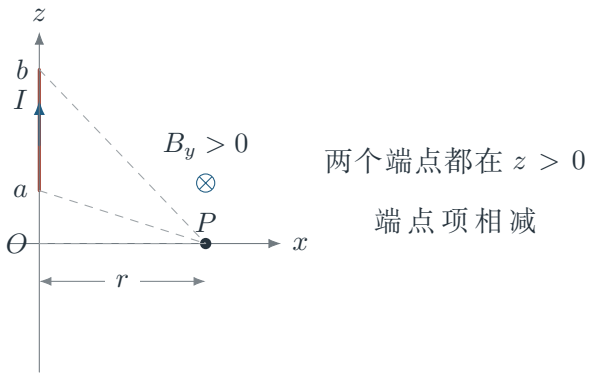


图 9-10　解答分析图　从每个源点到场点的位移与电流方向作叉乘

9-W2　螺线管中心场的近似误差

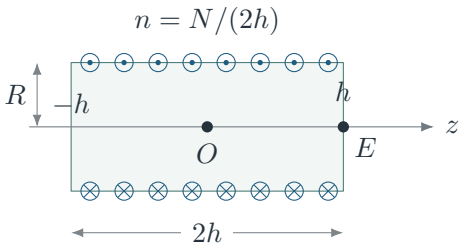
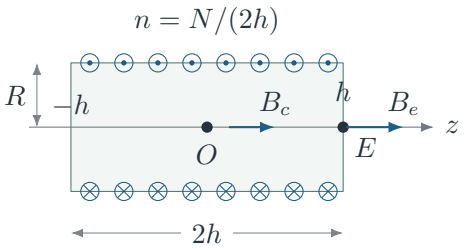


图 9-11　题图　有限螺线管的中心与端面轴心



$$B_c < \mu_0 n I$$

$$B_e/B_c \to 1/2$$

$$L/R \geq 14.04$$

图 9-12　解答分析图　中心场与端面场使用不同的端点距离

9-W3　线圈间距误差与二阶起伏

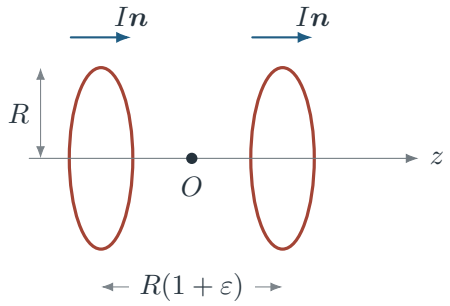
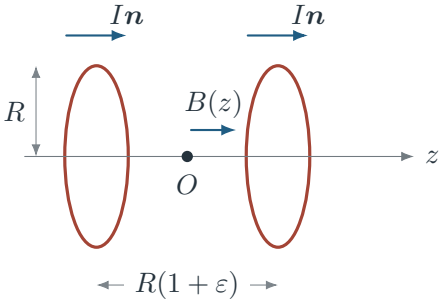


图 9-13　题图　同向线圈的实际间距偏离半径



$$\varepsilon > 0: \quad B''(0) > 0$$

$$\varepsilon < 0: \quad B''(0) < 0$$

图 9-14　解答分析图　中点二阶曲率的符号随间距误差改变

9-W4　反向线圈的中心梯度

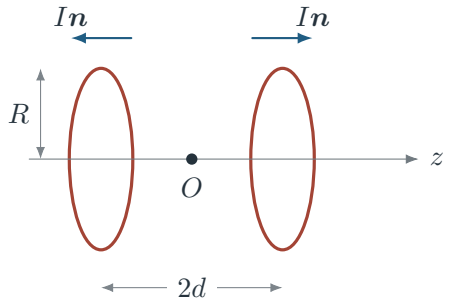
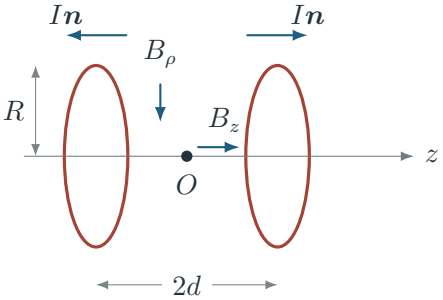


图 9-15　题图　反向线圈，正 z 端磁矩沿正轴向



$g > 0$

$2d = R$ 时 g 最大

横向梯度 $= -g/2$

图 9-16　解答分析图　中心附近轴向场外向，横向场内向

9-W5　多次穿面的代数电流

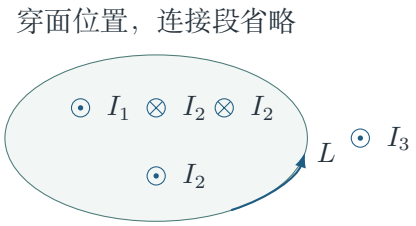


图 9-17　题图　以圆盘法向为正，点表示正穿越，叉表示负穿越

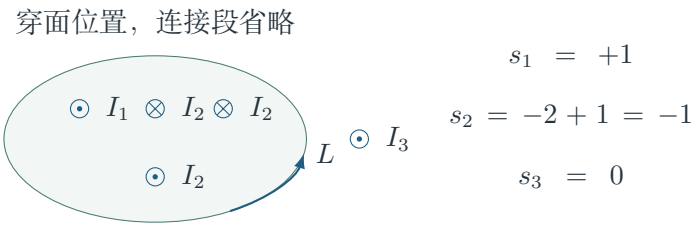
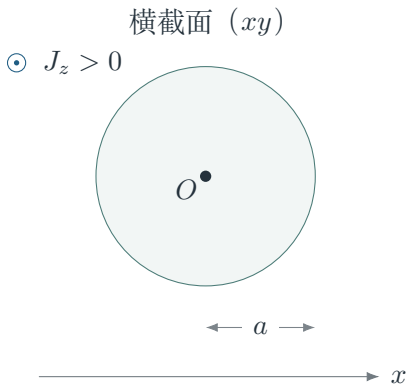


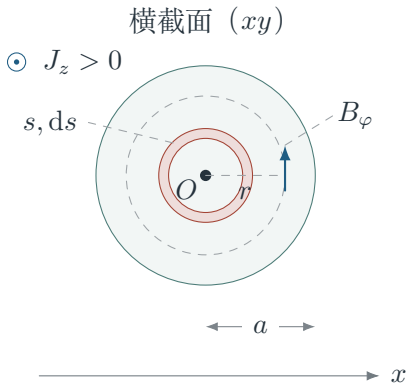
图 9-18　解答分析图　重复穿面按带符号次数计入

9-W6　径向增大的体电流密度



$$J_z = J_0 r^2 / a^2$$

图 9-19　题图　电流密度沿径向增大的圆柱截面



$$J_z = J_0 r^2 / a^2$$

$$I_{\mathrm{enc}} \propto r^4$$

$$B_{\mathrm{in}} \propto r^3$$

图 9-20　解答分析图　场的大小由圆盘累计电流除以周长决定

9-W7　空心导体的内外场

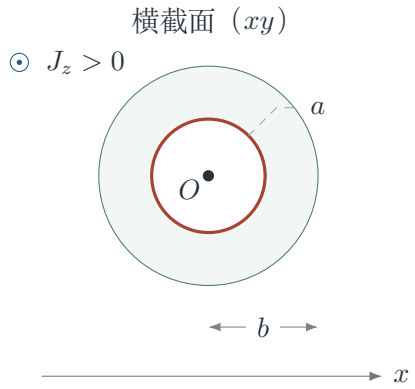


图 9-21　题图　完整同轴空心导体，金属环内电流均匀

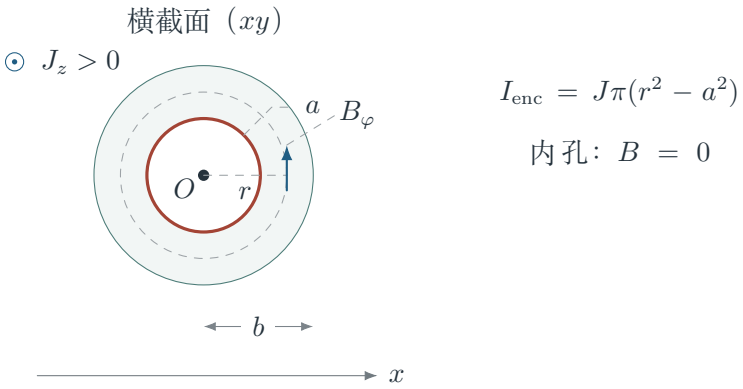


图 9-22　解答分析图　积分圆内只有金属部分承载电流

9-W8　同轴电缆的分段场

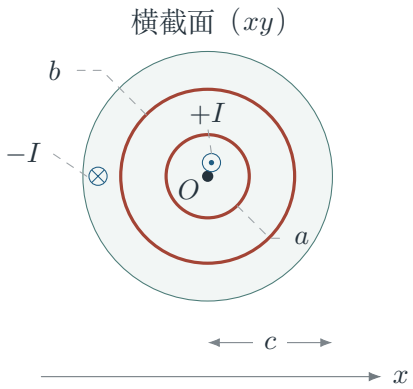
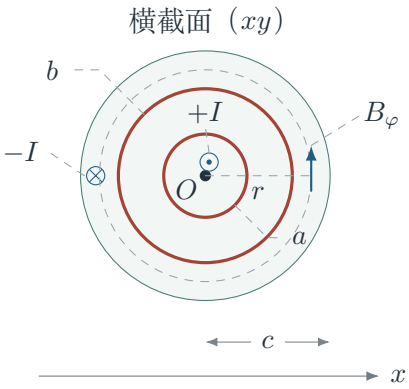


图 9-23　题图　内导体通入，外环形导体等量回流



I_{enc} 在回流层递减

$r \geq c:\ I_{\text{enc}} = 0$

图 9-24　解答分析图　同轴对称条件下，外区场为零

9-W9　均匀电流层

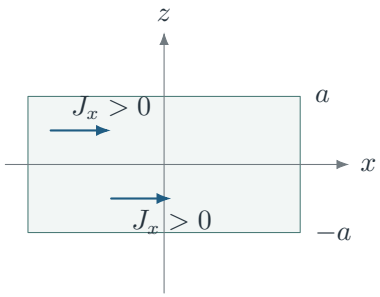


图 9-25　题图　电流沿正 x 方向，厚层关于中面对称

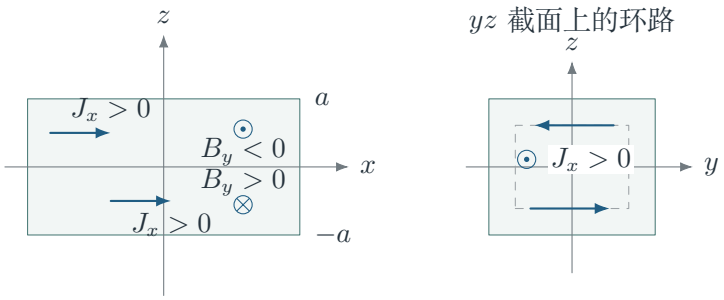


图 9-26　解答分析图　上半层场沿负 y，下半层场沿正 y

9-W10　环形线圈的径向起伏

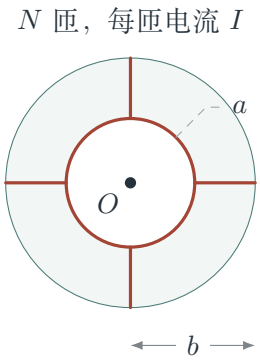
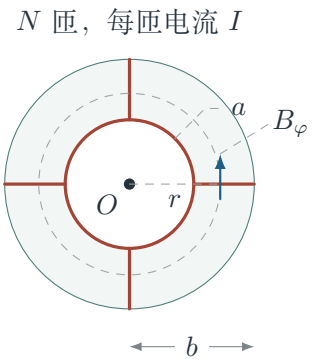


图 9-27　题图　环形线圈的内外半径与绕线示意



$$B_\varphi \propto 1/r$$

最大偏差在内侧 $r = a$

图 9-28　解答分析图　同心环路求出径向变化，不能假定截面内均匀

9-W11　偏心圆孔中的均匀场

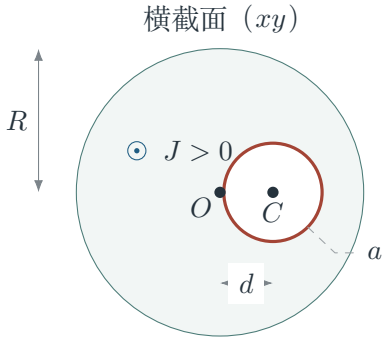
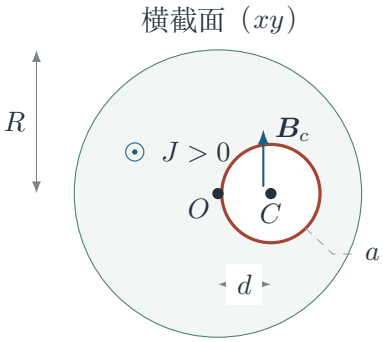


图 9-29　题图　偏心孔的轴线位移沿正 x



主 柱：$+\boldsymbol{J}$

孔 柱：$-\boldsymbol{J}$

$$\boldsymbol{B}_c = \mu_0 J d \boldsymbol{e}_y / 2$$

图 9-30　解答分析图　两个内部场的位置项消去，留下均匀孔场

9-W12　镜面上的磁场分量

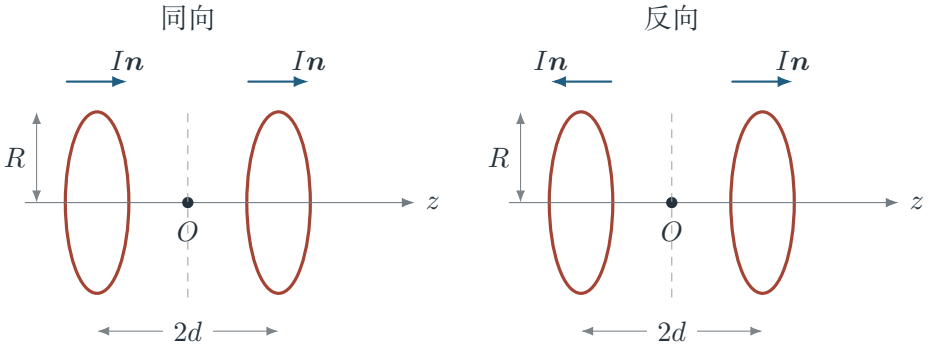


图 9-31　题图　两类电流配置的中面均为 $z=0$

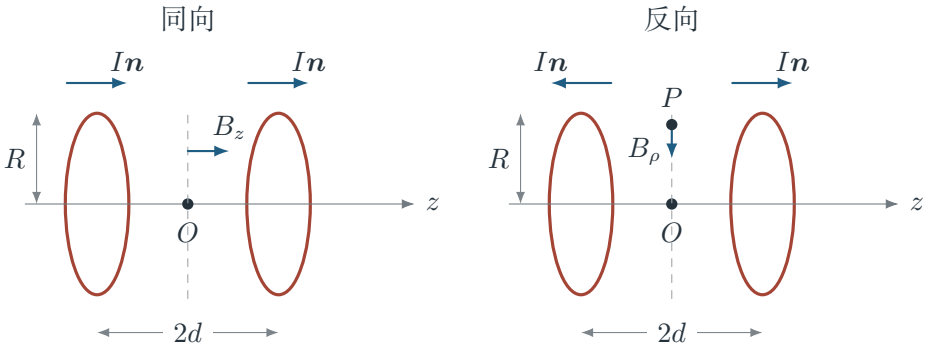


图 9-32　解答分析图　同向中面仅允许轴向场，反向中面仅允许径向场

9-T1　垂足两侧的有限导线

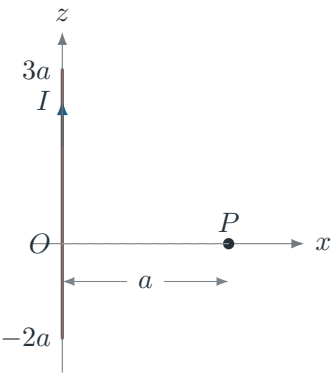


图 9-33　题图　垂足两侧长度不同的直线段

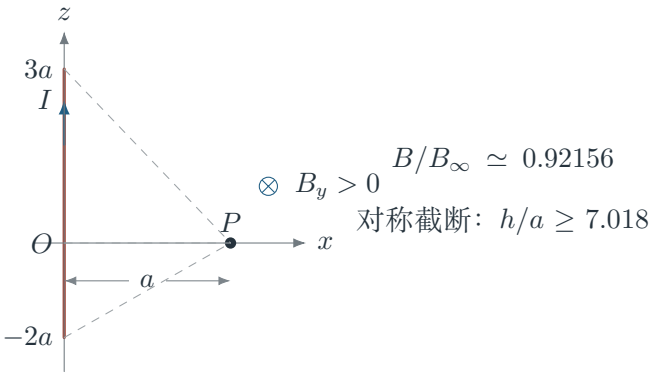


图 9-51　解答分析图　端点坐标一正一负，两个正贡献相加

9-T2　两半段匝密度不同的螺线管

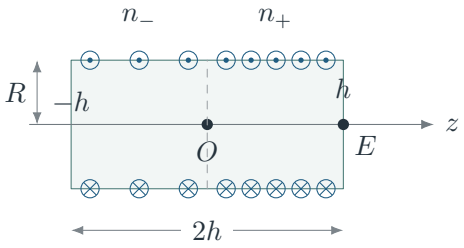
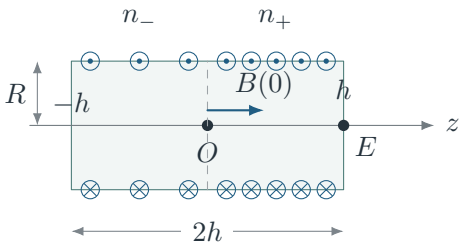


图 9-34　题图　两半段同向通电，绕线间距示意取 $n_+ > n_-$



$$B(0) \propto n_+ + n_-$$

$$B'(0) \propto n_+ - n_-$$

图 9-52　解答分析图　中心场取匝密度之和，中心梯度取两者之差

9-T3　亥姆霍兹线圈的离轴场

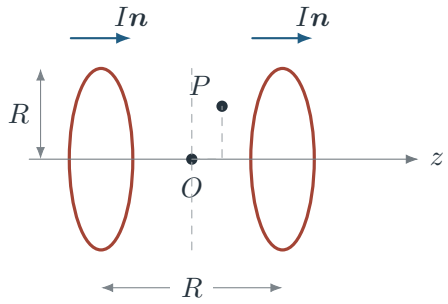
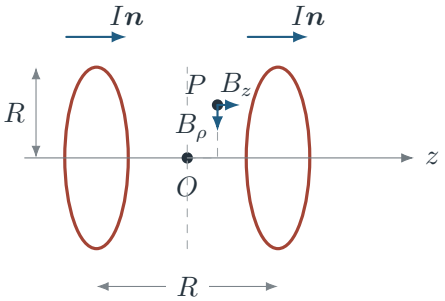


图 9-35　题图　中心附近离轴点的轴向和径向坐标



$$B_\rho(\rho,0) = 0$$

$$B_z(\rho,0) < B_H$$

图示点取 $\rho^2 > 4z^2/3$

图 9-53　解答分析图　径向场方向依位置改变，图示条件下指向轴线

9-T4　圆回路的磁标势分支

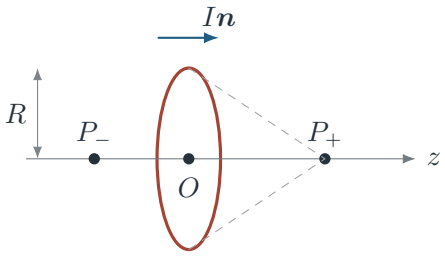
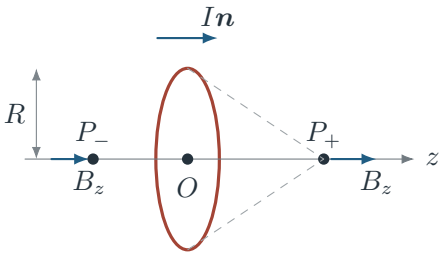


图 9-36　题图　圆回路两侧的轴上场点和圆盘支面



$\Omega(0^+) = +2\pi$

$\Omega(0^-) = -2\pi$

两侧 B_z 连续

图 9-54　解答分析图　立体角有换支跳跃，磁场方向在两侧相同

9-T5　套连数与环流

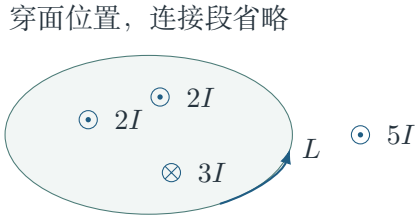


图 9-37　题图　电流回路的带符号套连数

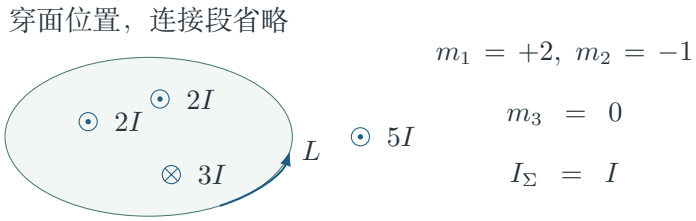
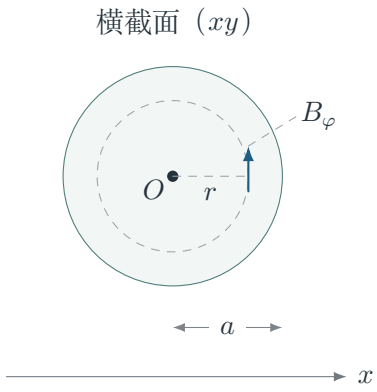


图 9-55　解答分析图　积分曲线不跨过源导线时，套连数保持不变

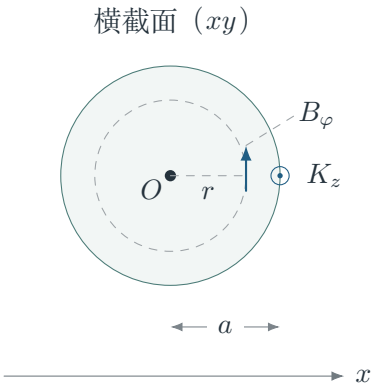
9-T6　由分段磁场反求电流



$$r < a:\ B_\varphi = \beta r$$

$$r > a:\ B_\varphi = \gamma / r$$

图 9-38　题图　给定分段场，边界可承载轴向面电流



$$r < a:\ B_\varphi = \beta r$$

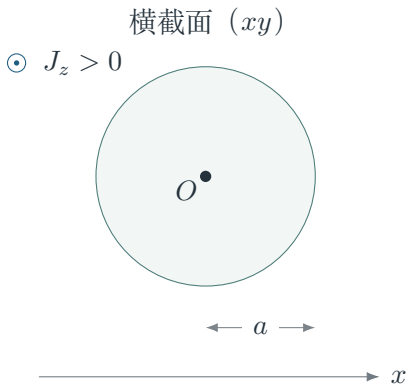
$$r > a:\ B_\varphi = \gamma / r$$

$$J_z = 2\beta/\mu_0$$

$$K_z = (\gamma/a - \beta a)/\mu_0$$

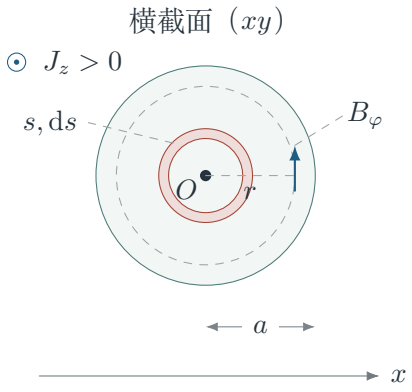
图 9-56　解答分析图　图中面电流箭头取 $\gamma > \beta a^2$，反号时电流反向

9-T7　中心集中的电流分布



$$J_z = J_0(1-r^2/a^2)$$

图 9-39　题图　电流密度在轴心最大、表面为零



$$J_z = J_0(1-r^2/a^2)$$

$$r_* = a\sqrt{2/3}$$

$$B_{\max} > B(a)$$

图 9-57　解答分析图　场峰值出现在导体内部，累计电流与周长共同决定峰值

9-T8　环路外直导线的局部场

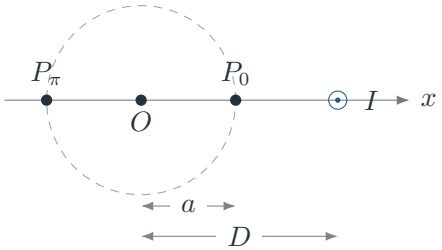
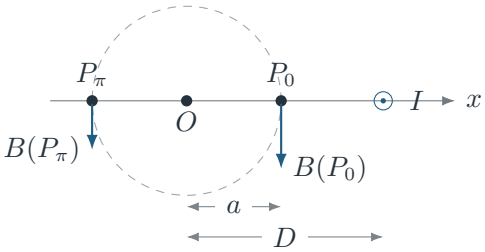


图 9-40　题图　外部直导线不穿过积分圆盘



$$B_t(0) < 0$$

$$B_t(\pi) > 0$$

$$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = 0$$

图 9-58　解答分析图　近点与远点的场同沿负 y，切向投影却相反

9-T9　两张平行电流面

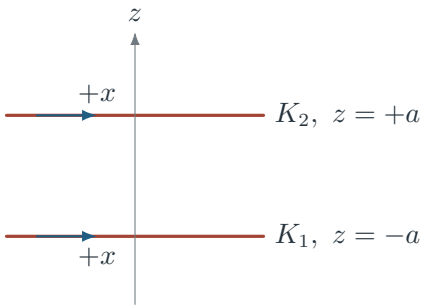


图 9-41　题图　两电流面，箭头标正 x，电流强度可带符号

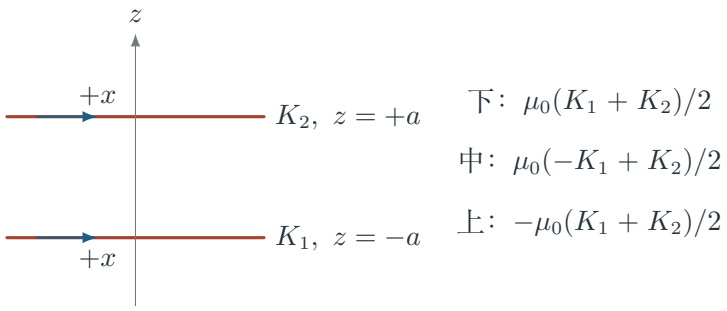


图 9-59　解答分析图　各区域叠加单张电流面的上下侧场，式中均为 B_y

9-T10　螺距不能忽略的圆柱面电流

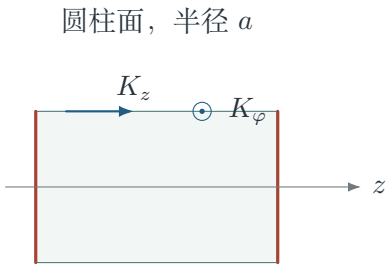


图 9-42　题图　轴向与环向面电流叠加的连续模型

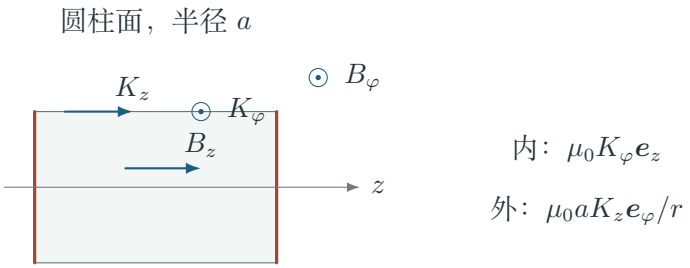


图 9-60　解答分析图　两个电流分量分别产生轴向内部场和环向外部场

9-T11　环形线圈截面的磁通

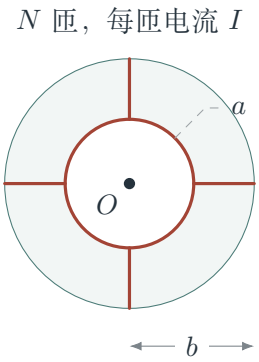


图 9-43　题图　矩形截面环形线圈

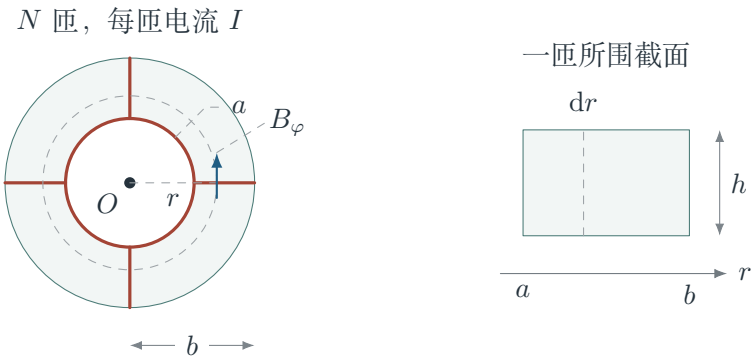


图 9-61　解答分析图　沿径向条带积分磁通，每条带上的场值不同

9-T12　反向体电流层

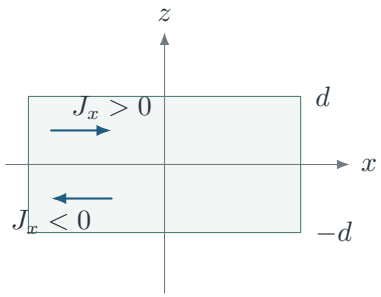


图 9-44　题图　上下半层电流反向，密度随 z 线性变化

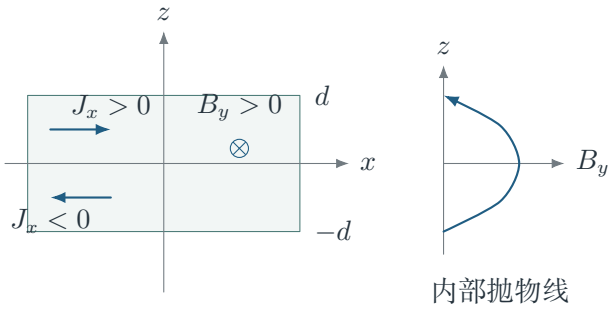


图 9-62　解答分析图　内部场沿正 y，边界归零，中面最大

9-T13　两个偏心空腔

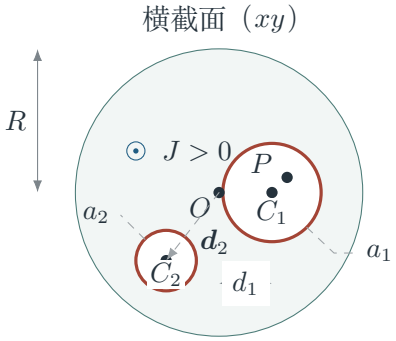


图 9-45　题图　两个互不相交的偏心圆孔

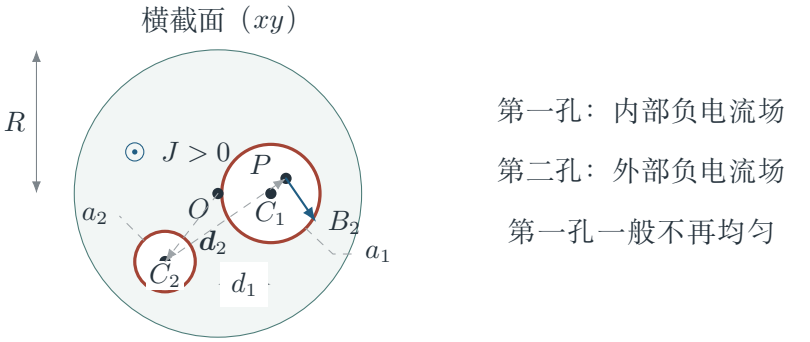


图 9-63　解答分析图　C_2 到 P 的箭头为 $\boldsymbol{s}_2$，第二个孔柱的外场随位置变化

9-T14　平面回路两侧的镜像场

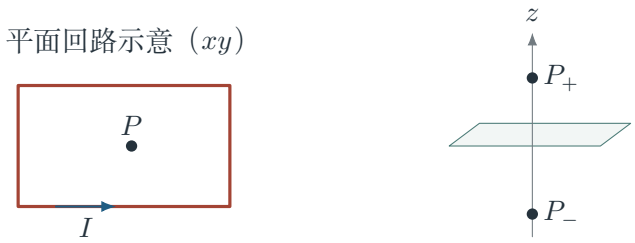


图 9-46　题图　左图为回路平面，右图为镜像场点示意

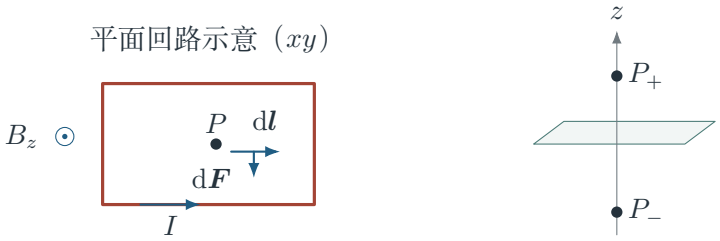


图 9-64　解答分析图　平面内非源点的磁场法向，平面内电流元的力切向；图取 $B_z > 0$

9-T15　斜截面穿过圆柱电流

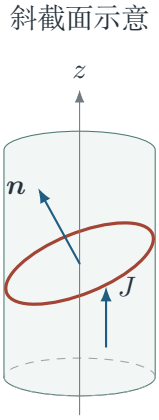
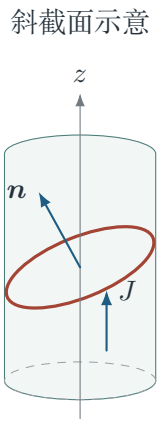


图 9-47　题图　圆柱电流与倾斜椭圆截面



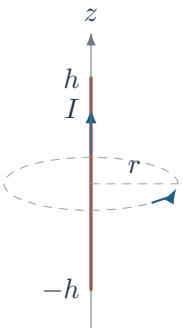
$$S_\Sigma \cos\theta = \pi a^2$$

$$J_n = J\cos\theta$$

$$I_\Sigma = J\pi a^2$$

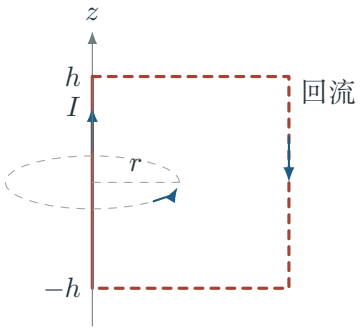
图 9-65　解答分析图　面积增大与法向电流分量减小恰好抵消

9-T16　孤立直线段与环路定理



同轴积分圆的透视图

图 9-48　题图　有限直线段的贡献不构成完整稳恒源



同轴积分圆的透视图

线段环流 $< \mu_0 I$

其余回路补足环流

图示回流不穿过圆盘

图 9-66　解答分析图　闭合连接与回流必须计入总场

9-T17　非均匀回流的同轴电缆（限时综合）

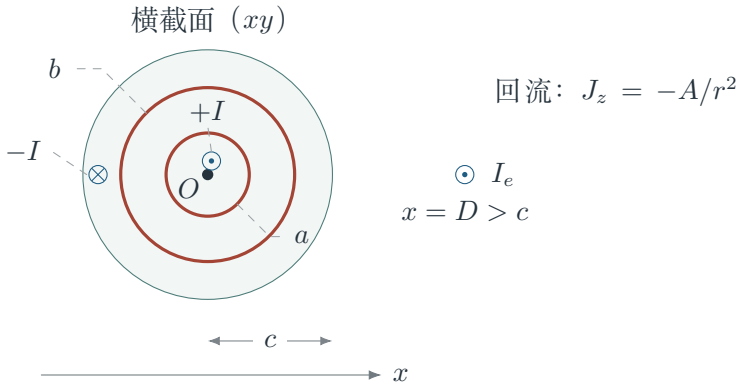


图 9-49　题图　同轴回流电缆与环路外平行导线

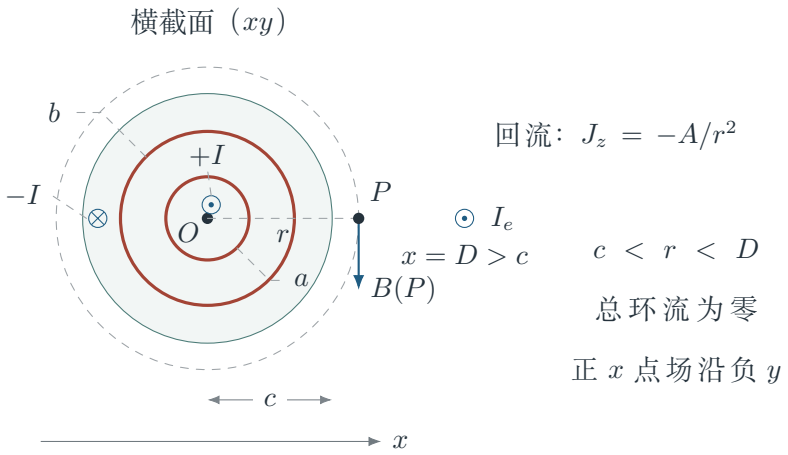


图 9-67　解答分析图　电缆外场消失，外部导线仍改变积分圆上的局部场

9-T18　偏心空腔内的载流细线（限时综合）

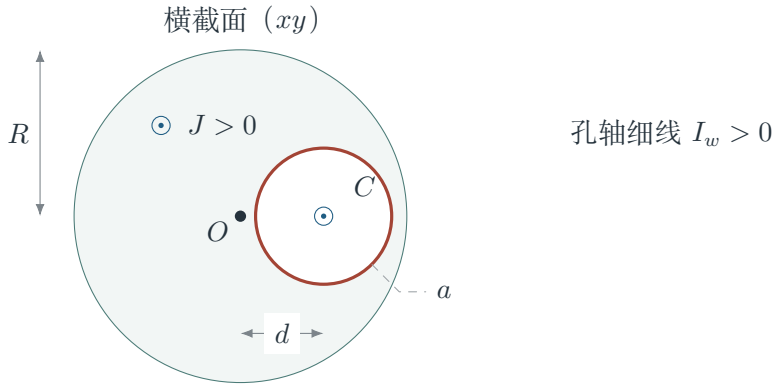


图 9-50　题图　孔轴细线与金属电流方向相同

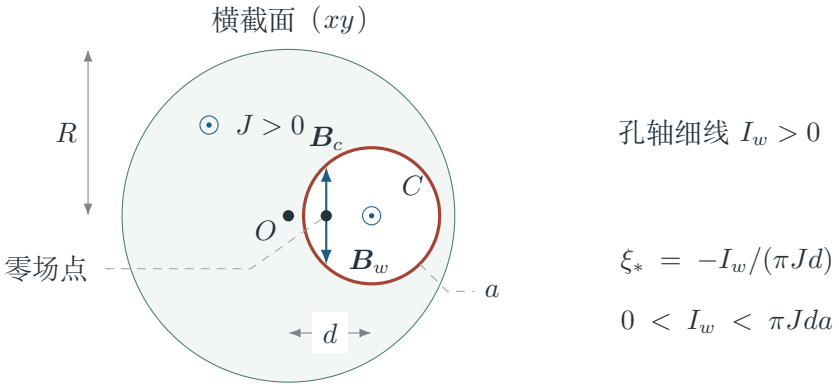


图 9-68　解答分析图　在孔轴左侧，两部分场反向；图示满足孔内零点条件